Project Design Phase

Proposed Solution

Date	17 July 2026
Team ID	
Project Name	SB Stocks — MERN Stack Stock Trading Simulation Platform
Maximum Marks	2 Marks

The project team has filled in the following proposed solution template for SB Stocks:

S.No.	Parameter	Description
1	**Problem Statement (Problem to be solved)**	New and aspiring investors need a safe, realistic environment to practice real-time stock trading and portfolio management without risking real money — existing brokerage platforms involve real financial risk, complex onboarding (KYC, funding), and a steep learning curve for beginners.
2	**Idea / Solution description**	SB Stocks is a MERN stack (MongoDB, Express.js, React, Node.js) web application that simulates real-time stock trading. Users start with a $100,000 virtual balance, view a live, auto-refreshing price feed for major listed stocks, place Buy/Sell orders, track their portfolio (holdings, returns, weighting), and review a complete, auditable order and wallet history — all without any real financial exposure.
3	**Novelty / Uniqueness**	Unlike static paper-trading spreadsheets or overly complex professional trading terminals, SB Stocks pairs a live-updating price feed with a clean, beginner-friendly dashboard — candlestick/trend/volume charts, a wallet funding simulator, top gainers/losers widgets, and a fully auditable order book — delivered as an accessible, browser-based experience built entirely on the open-source MERN stack.
4	**Social Impact / Customer Satisfaction**	Improves financial literacy by giving students, first-time investors, and hobbyist traders a hands-on, risk-free way to understand market behaviour and trading mechanics before committing real capital — reducing the chance of costly beginner mistakes once they do start trading with real money.
5	**Business Model (Revenue Model)**	Freemium model: free access to the core simulator (live prices, trading, portfolio tracking) to drive adoption and reach; optional premium tier for advanced analytics, longer historical charts, and multiple virtual portfolios; potential B2B licensing to colleges and finance-training programs for classroom use.
6	**Scalability of the Solution**	Built on a stateless Express/Node REST API with MongoDB (Mongoose ODM), the system can scale horizontally behind a load balancer as user count grows. The market-data layer is decoupled behind an API abstraction, so it can be swapped for a live provider and cached (e.g., Redis) to support far higher request volumes without re-architecting the core application.